

Qatar-1b

R-1608

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_211107 · created 2026-08-02T03:05:57+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459525.716 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.016
Transit depth (Rp/Rs)^2	1.02 +/- 0.33 [%]
Orbital Inclination (inc)	89.07 +/- 1.75
Airmass coefficient 1 (a1)	1.075 +/- 0.015
Airmass coefficient 2 (a2)	-0.0386 +/- 0.005
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#1 - [396, 69]
Optimal Method	PSF photometry
Transit Duration (day)	0.0784 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.101 $\pm$ 0.016 vs 0.1463 (deviation 2.53 σ)
Duration fit vs published	0.0784 d vs 0.0692 d (deviation 3.3 σ)
Mid-time O-C	-5.5 min
Depth SNR	5.6
Residual scatter	1.39 %

Light curve

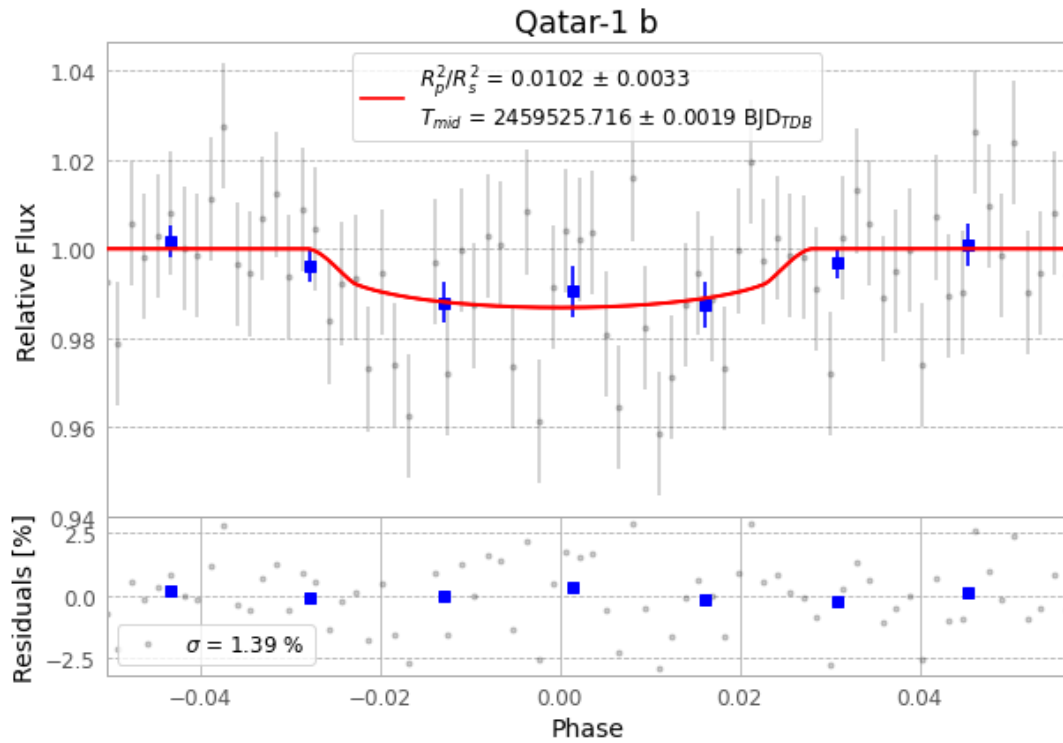


Figure 1 — FinalLightCurve_Qatar-1 b_2021-11-06.png (SHA-256 40a4b0744633a278...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ae67af4f580719da...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), Qatar-1211107032443.FITS → Qatar-1211107070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-211107.log (260a7571487e...), FinalLightCurve_Qatar-1 b_2021-11-06.png (40a4b0744633...), FinalLightCurve_Qatar-1 b_2021-11-06.csv (d5beb13f355d...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:05Z.